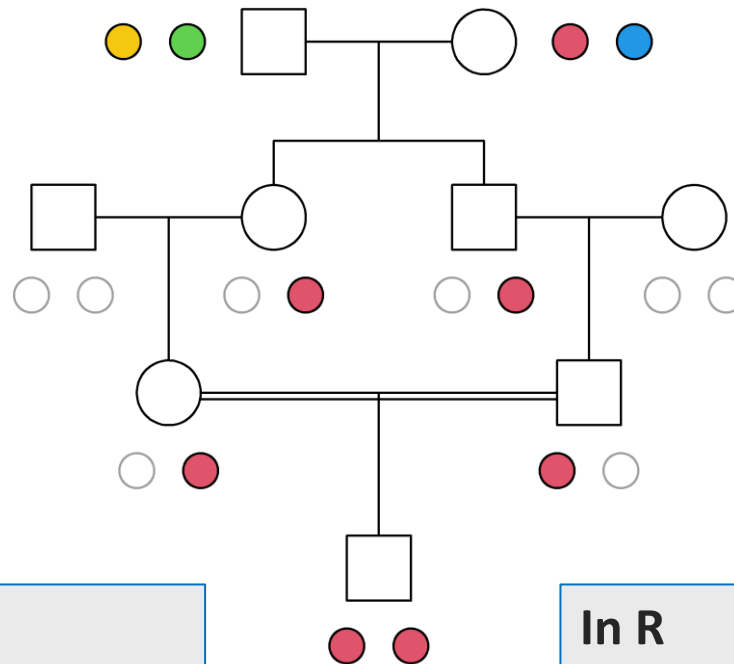


Inbreeding coefficient: Example

Wright's path formula:

$$\varphi_{P,Q} = \sum_A \sum_v \left(\frac{1}{2}\right)^{|v|+1} (1 + f_A)$$



By hand

$$\begin{aligned} f &= P(\text{red autozygous}) \cdot 4 \\ &= 0.5^6 \cdot 4 \\ &= 1/16 \end{aligned}$$

↑
other colors

In R

```
> library(pedsuite)

> x = cousinPed(1, child = T)
> inbreeding(x, ids = 9)

[1] 0.0625
```